Lecture 2: Planet Formation 101 & Orbital Dynamics

Planetary Systems

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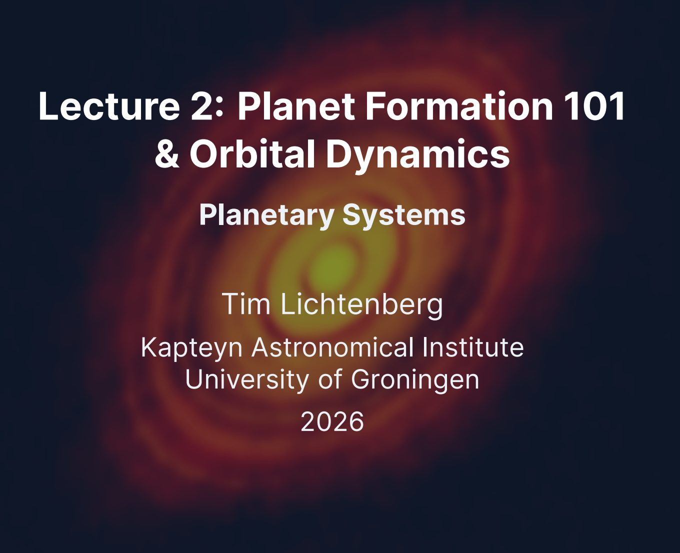
2026



HL Tauri, ALMA (2014). Credit: ALMA (ESO/NAOJ/NRAO)

Learning objectives

By the end of this lecture, you will be able to:

- Describe the stages of planet formation, from molecular cloud collapse to assembled planets.
- Explain how dust overcomes the growth barriers to become planetesimals, and how planetesimals become planets.
- Apply Kepler's laws and the vis-viva equation to quantitative orbital problems.
- Explain how resonances, tides, and migration reshape a planetary system after it forms.

Planet formation is a race: solids must assemble into cores before the gas disk disperses, and dynamics keeps rearranging the result long afterwards.

Recap: Lecture 1

Where we ended up last time:

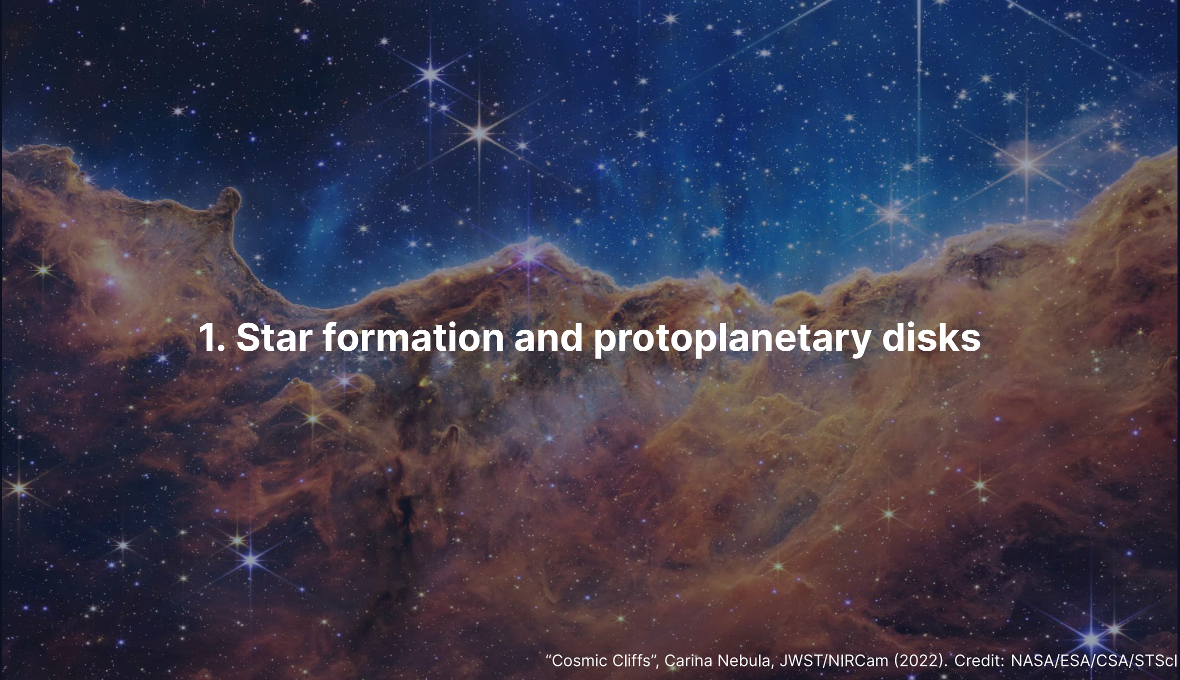
- Planetary science is **comparative**: planets as a set of natural experiments.
- Three driving questions: formation, habitability, life.
- The solar system has **structure**: rocky inside, gas and ice giants outside.
- The Sun holds **99.86%** of the system's mass; Jupiter holds 71% of the rest.

Our one derivation so far

$$M_* = \frac{4\pi^2 r^3}{GP^2}$$

Newton's form of Kepler's third law, for a *circular* orbit.

Today: where that structure came from, and the general orbital tools that replace the circular-orbit shortcut.



1. Star formation and protoplanetary disks

“Cosmic Cliffs”, Carina Nebula, JWST/NIRCam (2022). Credit: NASA/ESA/CSA/STScI

Every planetary system starts in a cold cloud

Molecular clouds: $T \sim 10$–20 K, mostly H_2 and helium, with ~ 1% by mass in microscopic **dust grains**.

Collapse begins when self-gravity beats thermal and magnetic pressure support. The critical mass is the **Jeans mass**:

$$M_J \sim \frac{c_s^3}{\sqrt{G^3 \rho}}$$

Order of magnitude

For $T \approx 10$ K and $n_{\mathrm{H}_2} \approx 10^4\ \mathrm{cm}^{-3}$:

$$M_J \sim 1\text{–}10\ M_\odot$$

Not a coincidence: the Jeans mass under molecular-cloud conditions is *comparable to a stellar mass*. Clouds fragment into star-sized pieces.

Star formation, caught in the act

Credit: NASA/ESA/CSA/STScI (JWST/NIRCam)

The “Cosmic Cliffs” at the edge of the Carina Nebula.

- Embedded protostars and outflows along the eroding cloud edge.
- Ultraviolet radiation from young massive stars sculpts the gas.
- A present-day analogue of the environment the Solar System was born in.

Why collapse makes a disk, not a ball

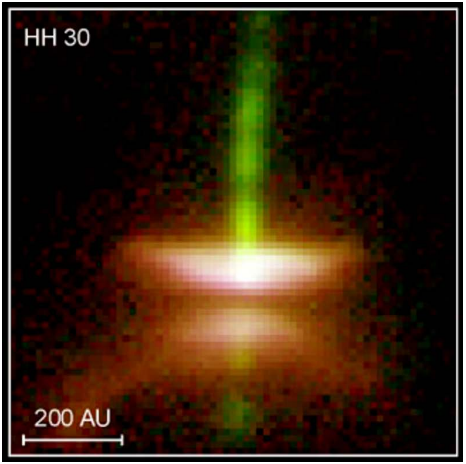
Angular momentum is the decisive constraint: even a slow initial rotation is amplified enormously by contraction. For a cloud core of radius $R_0 \sim 0.1$ pc collapsing to stellar scales ($\sim 1\, R_\odot$):

$$\frac{\Omega_{\mathrm{final}}}{\Omega_{\mathrm{initial}}} = \left(\frac{R_0}{R_\odot}\right)^2 \sim 10^{13}$$

- No star could sustain that spin: the material cannot all fall onto the protostar.
- Instead it settles into a rotating, flattened **protoplanetary disk**.
- The protostar accretes from it over ~ 1–3 Myr; the disk becomes the raw material for planets.

Planet formation is a by-product of the angular-momentum problem of star formation.

A disk seen edge-on: HH 30

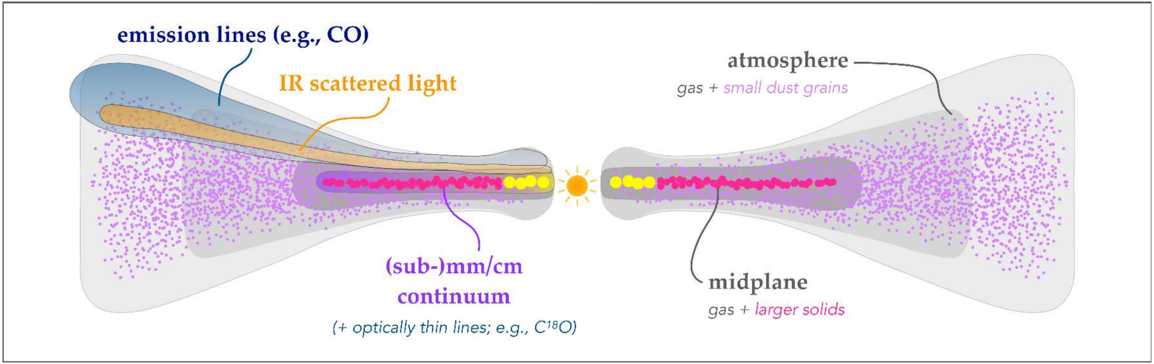


Credit: NASA/ESA/STScI (HST)

The geometry is directly visible:

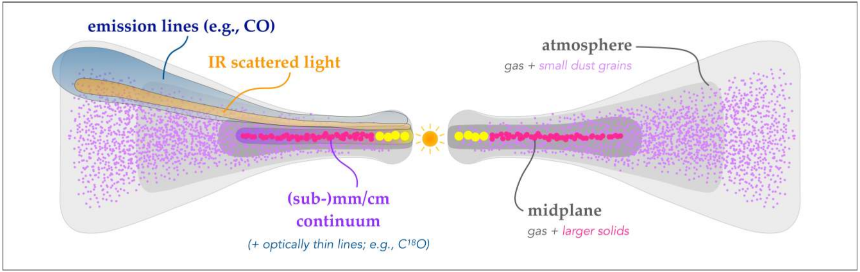
- **Dark lane**: the dusty midplane, opaque even in the optical.
- **Two bright lobes**: starlight scattering off the upper and lower disk surfaces.
- The star itself is hidden behind the midplane.

This is what “geometrically thin and flared” looks like on the sky.



Schematic cross-section of a protoplanetary disk. Adapted from Andrews (2020), Fig. 1.

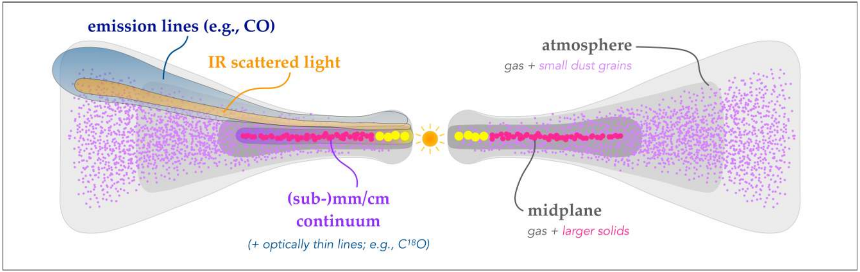
Reading the disk schematic



Gas in grayscale; solids drawn with *exaggerated* sizes.

- **Left half, what we observe**: emission lines (CO and others) from the warm surface; infrared scattered light from small grains in the atmosphere; (sub-)mm/cm continuum from large grains near the midplane.
- **Right half, the vocabulary**: a gas *atmosphere* holding small suspended grains, and a *midplane* layer of gas plus larger solids.

Reading the disk schematic



- **Flaring**: the disk gets vertically thicker, in relative terms, with distance.
- **Settling**: bigger particles sink toward the midplane, so different wavelengths trace different heights *and* different particle sizes.

Every observation of a disk is a slice through this structure: know which slice before interpreting it.

Disk structure in four numbers

- **Temperature gradient**: > 1500 K in the innermost disk, where silicates vaporise, falling to < 30 K in the outer disk. This profile decides what can condense where.
- **Surface density**: $\Sigma(r) \propto r^{-p}$ with $p \approx 0.5$–1.5. Densest in the inner disk, but the *mass budget* is dominated by the far larger area outside.
- **Disk mass**: from a few percent of the stellar mass in the youngest embedded systems, down to ~ 0.1% by the Class II stage. These estimates carry large systematic uncertainty.
- **Dust-to-gas ratio**: ~ 1% inherited from the interstellar medium, but locally enhanced by settling and radial drift, which is what makes planet formation possible.

Vertical structure: scale height $H/r \sim 0.03$–0.1, increasing outward (a **flared** disk).

Andrews (2020); Miotello et al. (2023).

The snow line

The distance beyond which water condenses as ice. In the solar nebula: roughly **2.5–3.5 AU**, between Mars and Jupiter.

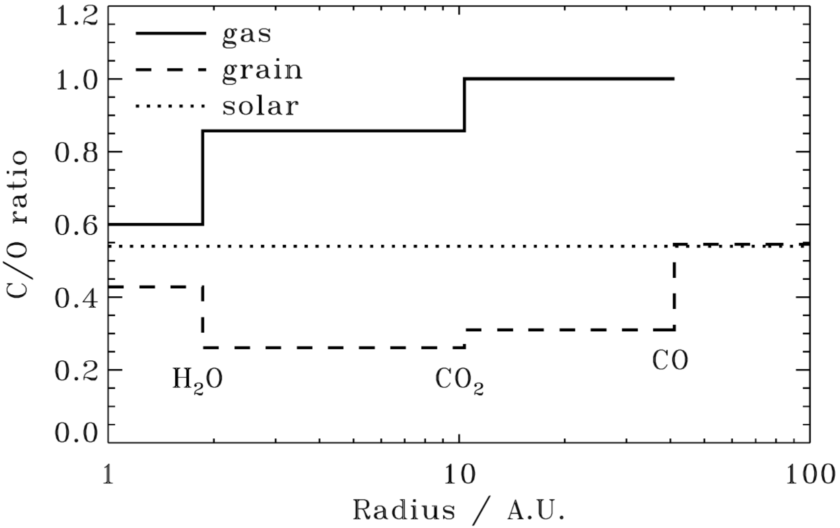
Why quoted values differ: *the snow line moves*.

- Young disk, strong accretion heating: out at ~ 5 AU.
- As accretion declines and the disk cools, it migrates **inward**.
- The 2.5–3.5 AU range is the location *imprinted on the asteroid belt's* compositional gradient (L4, L12).

Why it matters

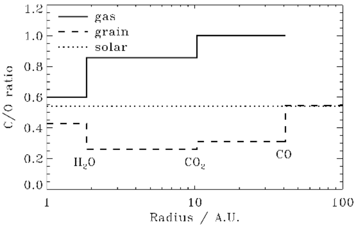
1. Adding water ice to the rock-and-metal inventory roughly **triples** the solid mass available.
2. The **CHNOPS** elements partition between volatile and refractory phases by local temperature, so *where* a planet forms sets its initial volatile budget.

Drążkowska et al. (2023); Lichtenberg et al. (2021); Krijt et al. (2023).



Sequential volatile condensation and the resulting C/O ratio across a disk. Reproduced from Öberg et al. (2011), Fig. 1.

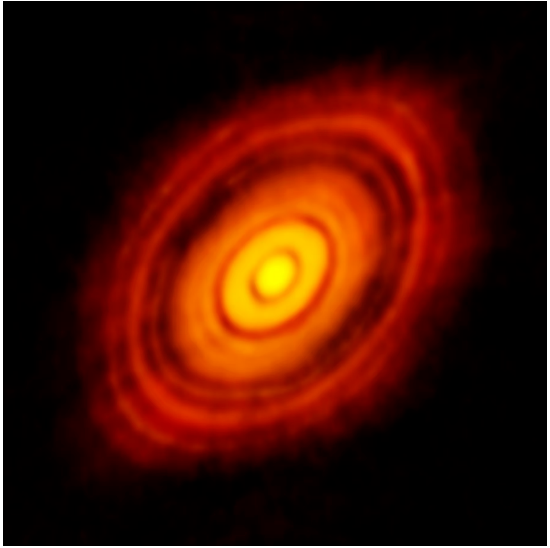
Reading the ice-line sequence



Solid curve: C/O of the *gas*. Dashed: C/O of the *grains*. Dotted: the stellar value, solar C/O = 0.54.

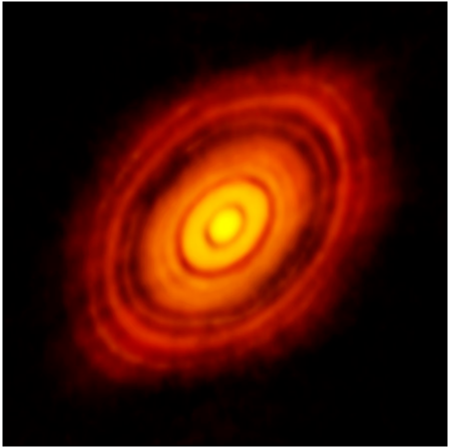
- H_2O ice line at ~ 2 AU: freezing out water *lowers* grain C/O and *raises* gas C/O.
- Beyond the CO_2 ice line at ~ 10 AU the gas reaches C/O = 1 and holds it out to the CO line at ~ 40 AU.
- Beyond the CO line essentially all C and O are frozen onto grains, so grain C/O returns to the stellar value.
- The exact H_2O line position is model-dependent, ~ 2–3.5 AU depending on the assumed temperature profile.

A planet inherits the chemistry of its formation zone. Composition is a *location* record, and we will use it as one in L12 and L13.



HL Tauri at 1.3 mm, ALMA (2014). Credit: ALMA (ESO/NAOJ/NRAO),

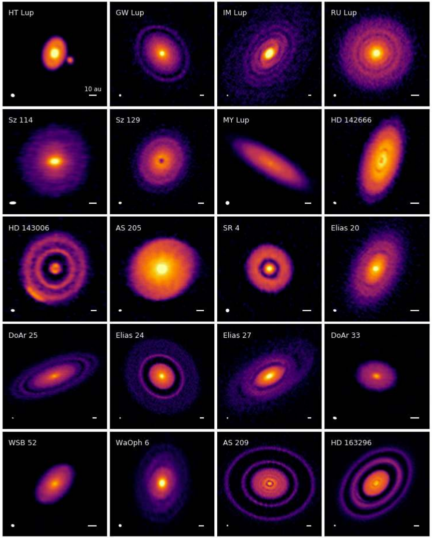
Reading HL Tau



The 2014 image that reset expectations for how early structure appears.

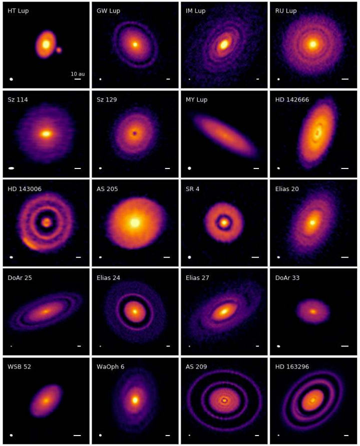
- Star age ~ 1 Myr: **young**, and already highly structured.
- Concentric bright rings and dark gaps out to ~ 100 AU, resolved at AU scales.
- Gaps *may* be carved by forming planets, but ice lines and magnetically driven structures remain viable alternatives.
- Either way: the dust is not smoothly distributed, and the conditions for planet formation are set early.

ALMA Partnership et al. (2015).



The 20 disks of the ALMA DSHARP survey at 240 GHz. Reproduced from Andrews et al. (2018), Fig. 3.

Reading the DSHARP gallery



- 20 nearby disks at 240 GHz (1.25 mm), ~ 5 AU resolution.
- Beam sizes (lower left) and 10 AU scale bars (lower right) in each panel; the colour stretch is chosen to bring out faint substructure.
- Rings and gaps appear in *nearly every* disk, not just HL Tau.
- **Caveat that matters**: the sample was deliberately biased toward the largest and brightest disks. Not every disk need be this richly structured.

Substructure is the norm among bright disks, and it appears early. Whatever forms planets is already at work in the first Myr.

The deadline: disk lifetimes

Infrared-excess surveys of clusters at different ages measure the fraction of stars still holding an optically thick *inner* disk.

- ~ 80% at 1 Myr, falling to < 10% by ~ 6 Myr.
- Half of all stars lose their inner disks within ~ 3 Myr.
- Essentially all disks are gone by ~ 6 Myr.

Dispersal is driven by viscous accretion onto the star, photoevaporation by stellar UV and X-rays, and probably magnetohydrodynamic winds.

A hard deadline for gas-giant formation: no core massive enough before the gas goes, no giant planet. This single constraint drives most of the theory that follows.

Haisch, Lada & Lada (2001).

2. From dust to planetesimals

Protoplanetary disk, artist’s concept. Credit: ESO/L. Calçada, CC BY 4.0

Twelve orders of magnitude

The raw material is sub-micrometre dust: silicates, metal oxides, and, beyond the snow line, ices.

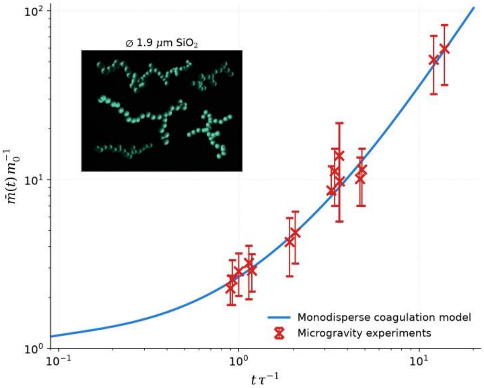
$$1\ \mu\mathrm{m} \longrightarrow 1000\ \mathrm{km} \qquad \text{is a factor} \sim 10^{12} \text{ in size.}$$

The physics changes character at almost every decade along the way:

- **Contact forces** (van der Waals) hold the smallest aggregates together.
- **Aerodynamics** (gas drag) dominates the pebble-to-boulder range.
- **Self-gravity** takes over only above roughly a kilometre.

There is no single mechanism that spans the whole range, which is why this section has several.

Grain growth: sticking and settling



Credit: Redrawn from Blum & Wurm (2008), Fig. 2

Grains collide through Brownian motion, turbulence, and settling.

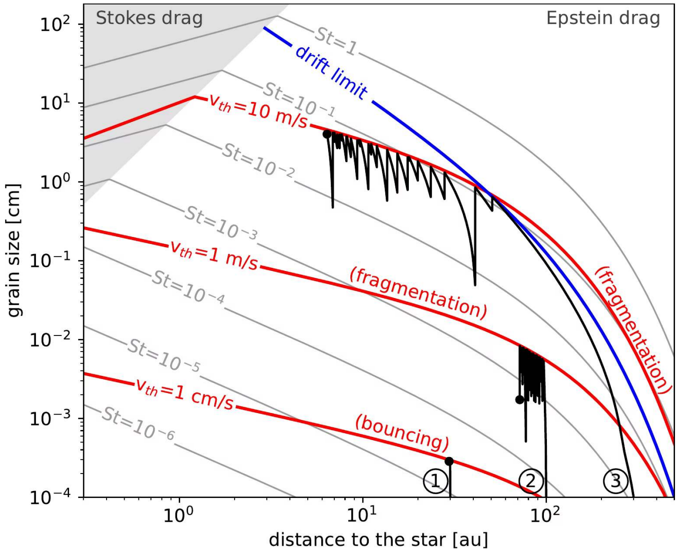
- At $\lesssim 1$ m s^{-1} they **stick**, held by van der Waals forces.
- The products are **fractal aggregates**: fluffy, porous, $D \approx 1.4$.
- **Inset**: 1.9 μm $\mathrm{SiO_2}$ monomers aggregated in microgravity.
- **Main panel**: aggregate mass vs time, in monomer units, against a Smoluchowski model.
- Efficient to mm and cm sizes. Then it stops.

Three barriers stop incremental growth

- **Bouncing barrier** (~ mm). Slightly faster collisions compact the aggregates, and compacted aggregates bounce instead of sticking.
- **Fragmentation barrier** (~ cm). Above ~ 1–10 m s^{-1}, depending on material, aggregates shatter. Silicates are fragile; ices survive faster impacts.
- **Radial drift barrier** (~ metre). Gas is partly pressure-supported, so it orbits slightly slower than Keplerian. Solids feel a **headwind**, lose angular momentum, and spiral inward. The drift rate *peaks* for metre-sized boulders, which reach the star in ~ 100–1000 years.

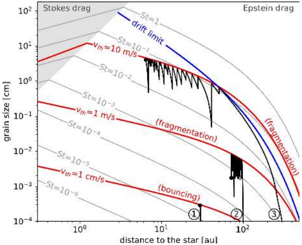
The metre-size barrier is the central problem: growth by sticking is slower than removal by drift, exactly in the size range you must cross.

Birnstiel et al. (2016).



Maximum grain size vs orbital distance in a standard disk model. Adapted from Drążkowska et al. (2023), Fig. 4.

Reading the growth-barrier map



Axes: orbital distance against the largest grain size that can survive there.

- **Red curves**: where collision speeds reach the bouncing (v_{th} = 1 cm s^{-1}) or fragmentation (1 and 10 m s^{-1}) thresholds.
- **Blue curve**: the radial-drift limit. Above it, grains drift inward faster than they grow.
- **Grey contours**: constant Stokes number, i.e. how tightly a particle couples to the gas.
- **Black curves**: representative growth histories, running into whichever limit is lowest.

Sticking stalls between tens of µm and dm; planetesimals are kilometres.

The gap is not marginal. Incremental sticking is off by many orders of magnitude, so a *collective* mechanism is required.

The streaming instability

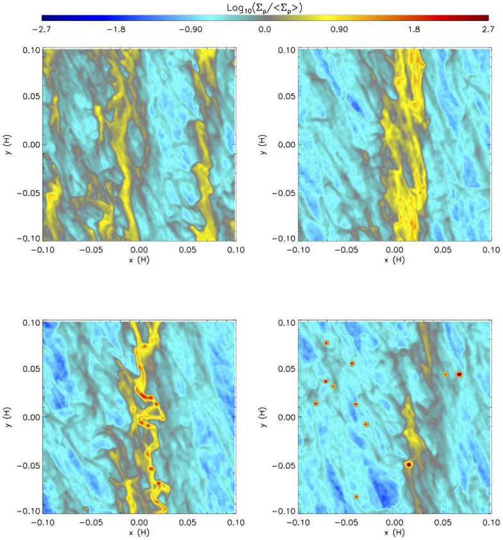
The way out is not stronger glue but **collective aerodynamics**.

When the local dust-to-gas ratio approaches unity, through settling to the midplane or accumulation at pressure bumps, the drag coupling changes character:

1. A local clump of particles **drags the gas along**, accelerating it toward the Keplerian velocity.
2. Faster gas means a **weaker headwind**, so the clump's own drift slows.
3. Particles drifting in from further out **catch up and join** the clump.
4. Positive feedback: dense filaments form, and where they are dense enough they **collapse under self-gravity**.

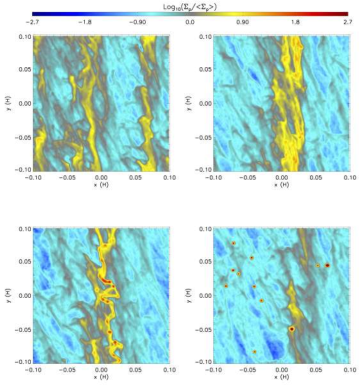
The product is a **planetesimal**, radius tens to a few hundred km, formed directly. The awkward metre-to-kilometre range is simply skipped.

Johansen et al. (2007); Simon et al. (2016).



Streaming-instability simulation: particle surface density in a local shearing box. Reproduced from Simon et al. (2016), Fig. 3.

Reading the streaming-instability snapshots

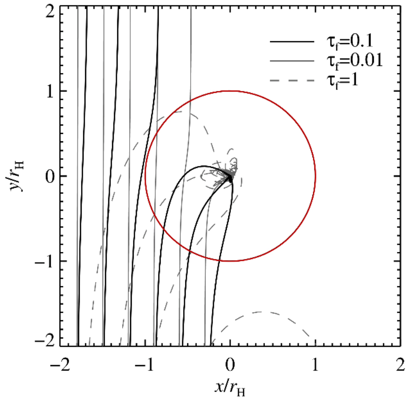


Four snapshots, time increasing from top left to bottom right. Colour: particle surface density normalised to its mean, logarithmic scale. Axes in gas scale heights *H*.

- **Early**: drag feedback has already organised the pebbles into loose, azimuthally extended **filaments**.
- **Then**: the filaments merge and sharpen into one dominant dense band.
- **Then**: the filaments fragment along their length under self-gravity.
- **Final panel**: gravitationally bound clumps (red points) have collapsed into planetesimals.

Note the scale: this is a *local* box, a few scale heights across, not a whole disk. The instability is a midplane process.

Pebble accretion



Credit: Lambrechts & Johansen (2012), Fig. 7

Large planetesimals sweep up the abundant mm- to cm-sized pebbles.

- Frame co-moving with the embryo; axes in Hill radii r_H; red circle is the Hill sphere.
- Line styles: friction times τ_f = 0.01, 0.1, 1, i.e. weaker gas coupling.
- Gas drag makes pebbles entering the Hill sphere **spiral in** instead of passing by.
- The effective cross-section far exceeds the geometric one.
- Cores reach ~ 10 $M_\oplus$ *within* the disk lifetime.

Pebble accretion switches itself off

Pebble accretion is **self-limiting**, and how it stops matters as much as how it works.

Pebble isolation mass

A sufficiently massive planet raises a **pressure bump** in the gas just outside its orbit. Inward-drifting pebbles pile up there and stop: the planet's supply is cut off.

The consequence is not local:

- The supply to the *entire disk interior to that orbit* is cut off at the same time, so one planet changes the solid budget everywhere inside it.
- A leading explanation for the isotopic split between inner and outer solar-system material (L12).

Drążkowska et al. (2023).

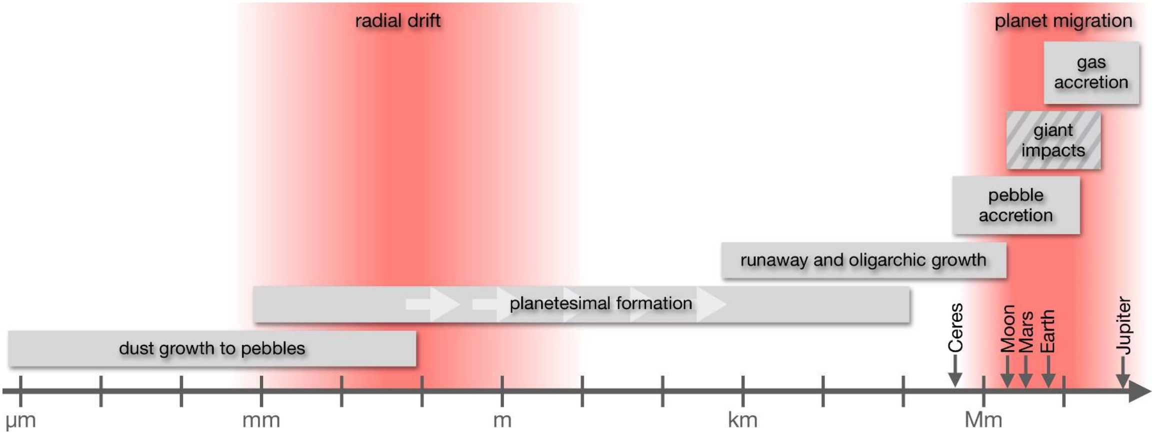
Planets do not form everywhere at once

Formation concentrates where solids concentrate:

- **Pressure bumps**: a local gas-pressure maximum halts radial drift, so solids accumulate instead of passing through.
- **Ice lines**: sublimation and recondensation enhance the local solid density and change grain stickiness.
- **Dead-zone edges**: boundaries between turbulent and quiescent regions trap material.

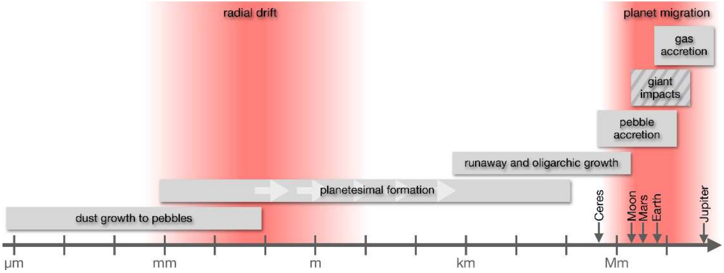
A non-uniform distribution of formation sites is one reason exoplanetary architectures are so diverse (L13).

Drążkowska et al. (2023).



Roadmap of the planet-formation sequence, organised by body size. Adapted from Drążkowska et al. (2023).

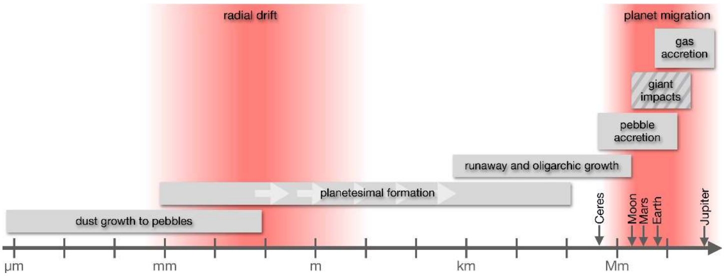
Reading the formation roadmap



Horizontal axis: body size, from sub-µm dust to gas giants. Reference bodies (Ceres, Moon, Mars, Earth, Jupiter) mark the scale.

- **Grey bars**: the size range over which each process operates, in order: dust growth to pebbles, planetesimal formation, runaway and oligarchic growth, pebble accretion, giant impacts, gas accretion.
- **Hatched bar**: giant impacts, the one process that *continues after the gas disk has dispersed*.

Reading the formation roadmap



- **Red bands**: the two great obstacles. Radial drift at pebble-to-boulder sizes, and planet migration at planetary masses.

The formation half of this lecture climbs this size axis from left to right; the right-hand red band, migration, is where we end today.

3. Building planets: runaway growth to gas giants

Giant impact, artist’s concept. Credit: NASA/JPL-Caltech

Runaway growth

Once planetesimals exist, **gravity** takes over from aerodynamics. A larger body bends incoming trajectories toward itself, so it collides with far more than its geometric cross-section:

$$\sigma_{\mathrm{eff}} = \pi R^2 \left(1 + \frac{v_{\mathrm{esc}}^2}{v_\infty^2}\right), \qquad v_{\mathrm{esc}} = \sqrt{\frac{2GM}{R}}$$

- In a dynamically cold swarm ($v_\infty \ll v_{\mathrm{esc}}$) the **focusing factor** is enormous.
- v_{esc} grows with mass, so *the biggest body grows fastest*: the size gap widens with time.
- At 1 AU, a single body reaches roughly **lunar mass within** ~ 10^5 **years**.

Growth toward Mars-mass embryos continues more slowly; the final terrestrial planets take far longer still (L4).

Kokubo & Ida (1998).

Oligarchic growth

Runaway growth destroys its own precondition.

- The growing bodies **stir up** the surrounding planetesimals, raising v_∞.
- Higher v_∞ shrinks the focusing factor, so the runaway slows into the **oligarchic regime**.
- A few dozen comparable-mass **oligarchs** each dominate a feeding zone, separated by ~ 5–10 mutual Hill radii.
- Each grows until it reaches its **isolation mass**: all material in its feeding zone consumed.

The problem this leaves behind

In the inner solar system the isolation mass is only ~ 0.01–0.1 $M_\oplus$, an order of magnitude short of Earth. Something later must merge the oligarchs.

Kokubo & Ida (1998).

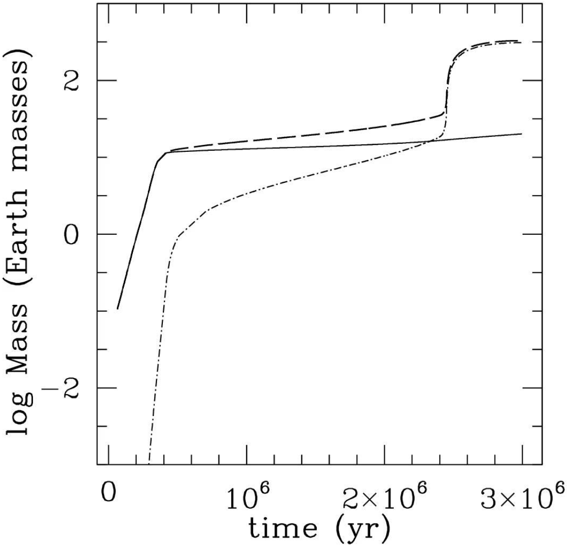
Core accretion: how to build a giant

The dominant model for giant-planet formation, in four steps:

1. A **solid core** grows from planetesimals and pebbles *beyond the snow line*, where solids are abundant.
2. At a critical core mass of ~ 10–20 $M_\oplus$, gravity can bind a substantial **gas envelope**.
3. The envelope contracts and cools, admitting more gas: **runaway gas accretion**. Hundreds of Earth masses of H and He within ~ 10^5 years.
4. Accretion ends when the planet **opens a gap** or the disk **dissipates**.

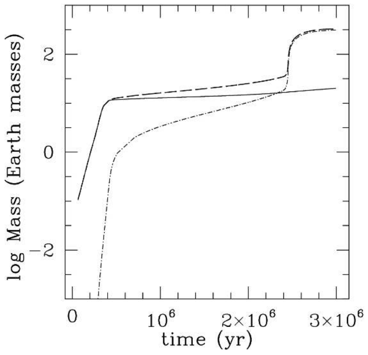
This explains the solar system's giants in one sentence: Jupiter and Saturn reached critical core mass before dispersal; Uranus and Neptune reached it too late, after most gas was gone.

Pollack et al. (1996).



Growth of a giant planet by core accretion at 5.2 AU. Reproduced from Helled et al. (2014), Fig. 1; simulation from Lissauer et al. (2009).

Reading the core-accretion curves



Mass against time for a classic simulation of Jupiter's formation at 5.2 AU.

- **Solid line**: core mass. **Dash-dotted**: envelope mass. **Dashed**: total.
- **Phase 1**: rapid core growth to ~ 10 $M_\oplus$.
- **Phase 2**: a long plateau of slow, quasi-hydrostatic envelope growth. This phase sets the clock, and it is what must fit inside the disk lifetime.
- **Phase 3**: runaway gas accretion, beginning at the **crossover point** where envelope mass overtakes core mass, at 2.3 Myr in this run, reaching a Jupiter mass by 3 Myr.

Compare 3 Myr here with the disk-lifetime plot: giant-planet formation only just fits inside the deadline.

Gravitational instability: the alternative pathway

A massive disk that cools rapidly enough can **fragment directly** into self-gravitating clumps, which contract into giant planets. No solid core needed first.

- Fast: orbital timescales, ~ 10^3 years.
- But it needs high disk mass *and* efficient cooling.

Boss (1997); Kratter & Lodato (2016).

Where those conditions hold

Only at large orbital distances, $\gtrsim$ 30–50 AU, in unusually massive disks.

So it is today regarded as a pathway for **massive companions on wide orbits**, not for the solar system's giants.

The giant impact phase

Credit: NASA/JPL-Caltech (artist's concept)

After the disk disperses, there is no gas drag left to damp eccentricities.

- Embryo orbits grow eccentric and start crossing.
- ~ 10–100 Myr of chaotic **giant impacts** merges the oligarchs into the final terrestrial planets.
- The **Moon-forming impact**, proto-Earth struck by the Mars-sized body Theia, is our most dramatic example.

This is the process that finally bridges the isolation-mass gap.

Formation is also a geophysical process

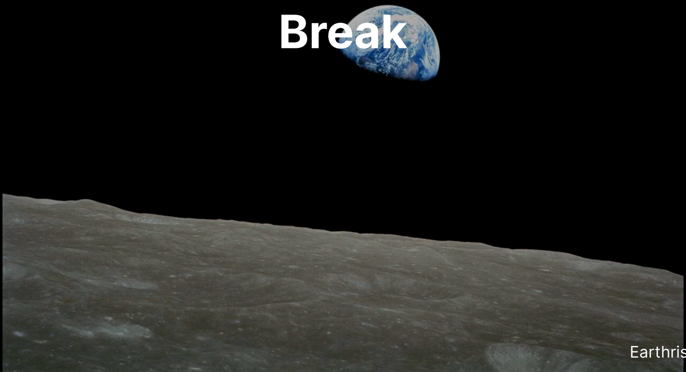
Assembly is not just bookkeeping of mass. The energy delivered changes what the planet *is*.

- Accretion and giant impacts **melt** the planet: a global **magma ocean**.
- Molten, dense metallic iron sinks and light silicates float: **core-mantle differentiation**.
- Short-lived radionuclides, above all $^{26}\mathrm{Al}$ (half-life 0.72 Myr), melt even *small planetesimals*, so differentiation can begin before assembly finishes.

The thermal and chemical state inherited from formation shapes everything downstream: magnetic field, volcanism, outgassing, habitability (L3, L4).

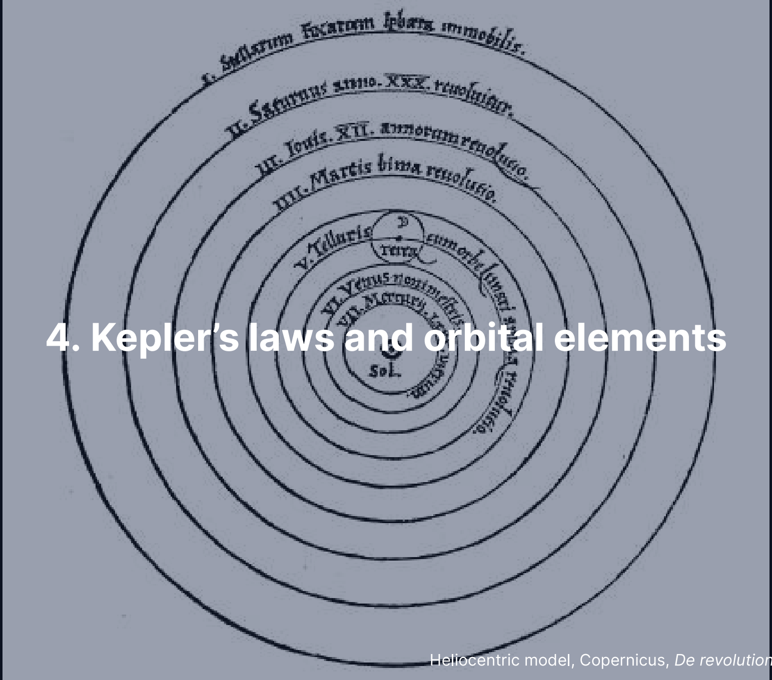
Lichtenberg et al. (2023).

Break



Earthrise, Apollo 8 (1968). Credit: NASA

4. Kepler’s laws and orbital elements



Heliocentric model, Copernicus, *De revolutionibus* (1543). Public domain

Kepler's three laws

Distilled by Kepler from Tycho Brahe's observations, early 17th century. Empirical first, explained later.

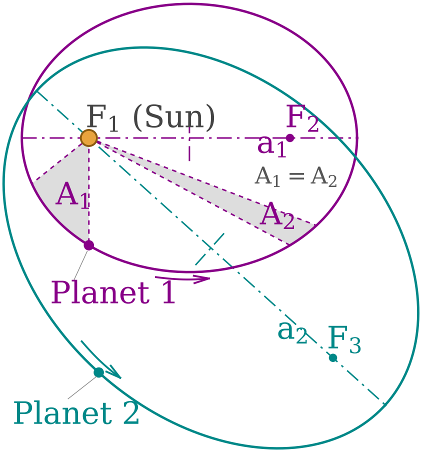
1. **The law of ellipses.** Each planet moves on an ellipse with the Sun at one *focus*, not the centre.
2. **The law of equal areas.** The Sun-planet line sweeps equal areas in equal times: fastest at perihelion, slowest at aphelion.
3. **The harmonic law.** $P^2 \propto a^3$, which in Newton's form reads

$$P^2 = \frac{4\pi^2}{G(M_1 + M_2)} a^3$$

Newton showed all three follow from one universal law of gravitation (L1).

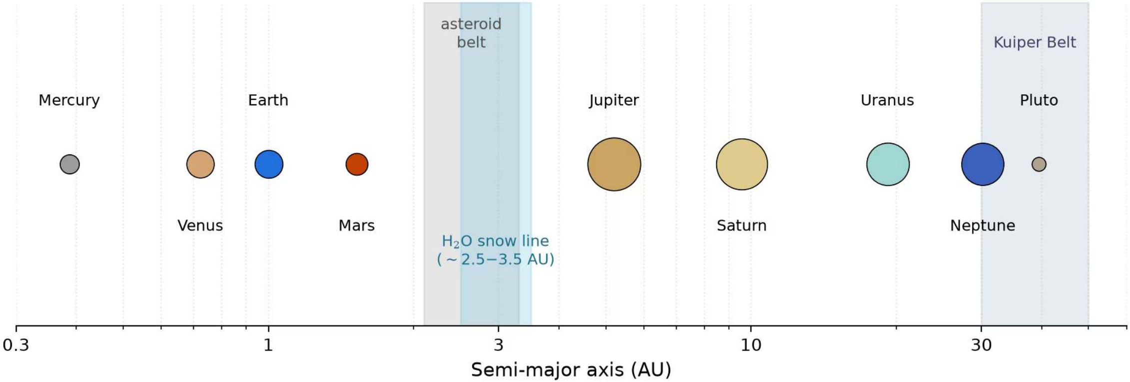
The third law is the workhorse of the whole course: it turns a *measured period* into a *mass*.

The three laws, geometrically



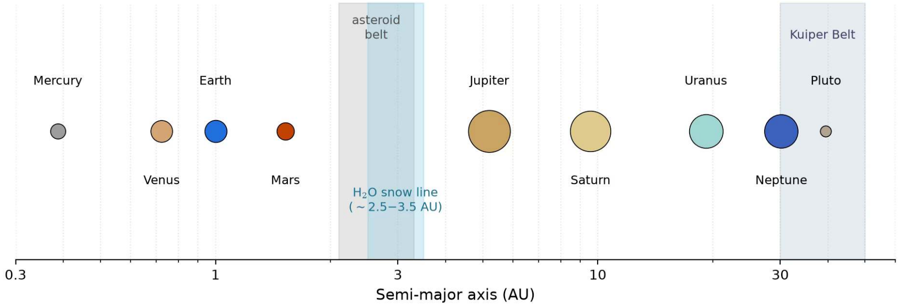
Credit: Hankwang, CC BY 2.5

- **(1)** The Sun sits at a focus. The empty focus has no physical meaning, which is why the ellipse was so hard to accept.
- **(2)** The shaded wedges have *equal area*, so they are swept in equal time: short and fat near perihelion, long and thin near aphelion.
- **(3)** $P^2 \propto a^3$ ties different planets' orbits to each other.
- Law 2 is angular-momentum conservation in disguise. We use it in the derivation shortly.



Architecture of the Solar System on a logarithmic semi-major-axis scale. Course-original figure; data: NSSDC Planetary Fact Sheet.

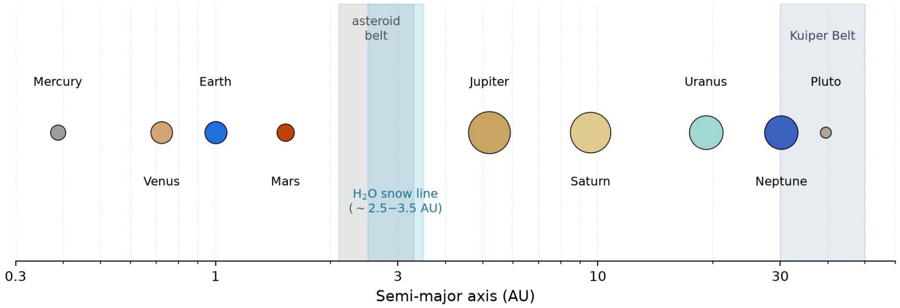
Reading the solar-system architecture



Logarithmic semi-major axis. Symbol areas scale as $\log(1 + R/R_\oplus)$ so that small bodies stay visible.

- **Main belt** shaded at 2.1–3.3 AU; **classical Kuiper Belt** at 30–50 AU.
- The 2.5–3.5 AU band marks where the solar nebula's **water snow line** lay: rocky planets inside, gas and ice giants outside.

Reading the solar-system architecture



- Note how much of the system is empty. The eight planets occupy a handful of decades in radius.
- This is the structure the first half of the lecture had to produce, and the second half will rearrange.

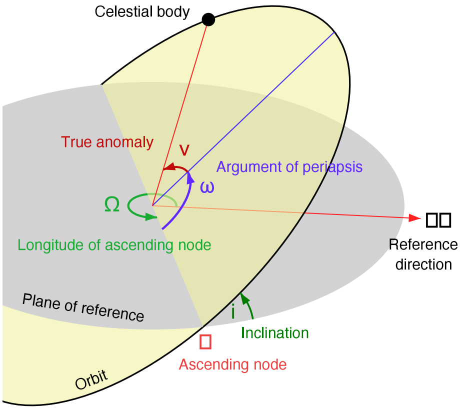
Six orbital elements

Element	Symbol	Meaning
Semi-major axis	a	Size of the orbit
Eccentricity	e	Shape ($e = 0$ circular, $0 < e < 1$ elliptical)
Inclination	i	Tilt of the orbital plane vs a reference plane
Longitude of ascending node	Ω	Where the orbit crosses the reference plane
Argument of perihelion	ω	Orientation of the ellipse within its plane
True anomaly	ν	Where the body is, right now

- The first five are **constant** for a Keplerian orbit: they define size, shape, and orientation.
- Only the sixth changes with time.

Perturbations, resonances, tides, and migration all act by *slowly changing the first five*. That is the subject of the rest of this lecture.

The four angular elements



Credit: Lasunncty, CC BY-SA 3.0

- **Grey**: the reference plane, usually the ecliptic for solar-system work.
- The orbital plane is tilted from it by the inclination i.
- Ω locates the **ascending node**, where the body crosses northward.
- ω rotates the ellipse *within* its plane (here "argument of periapsis", the general term).
- ν finally locates the body along the orbit.
- With a and e for size and shape, the description is complete.

What the solar system's elements tell us

Most planets have

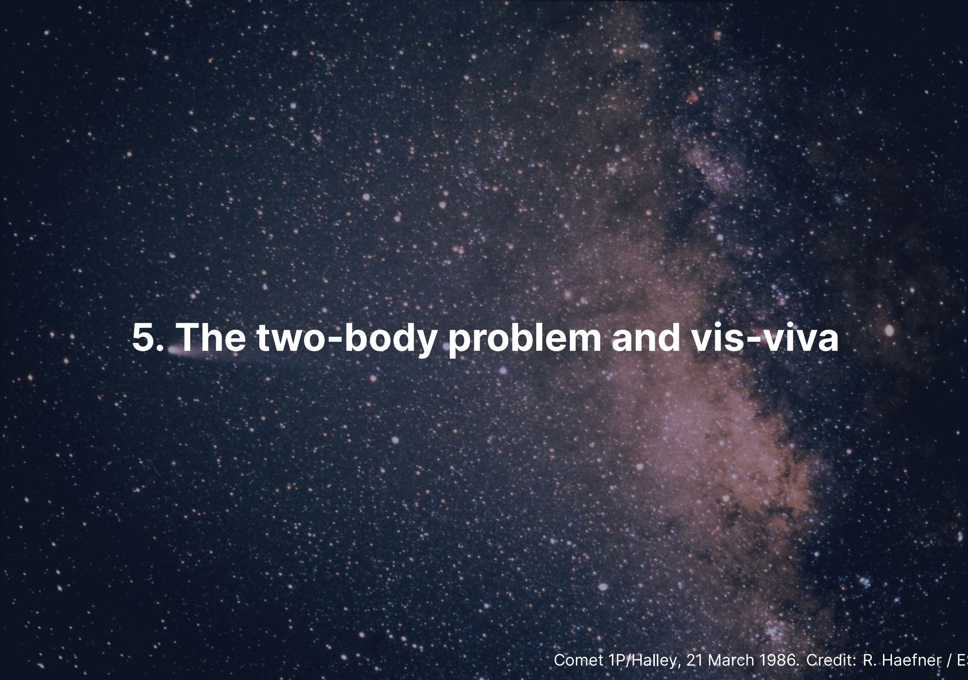
- nearly circular orbits, $e < 0.1$;
- low inclinations, $i \lesssim 3.5^\circ$.

That is not an accident: it is the **fossil signature of the disk** the planets formed in. A flat, cold disk makes flat, circular orbits.

The exceptions are the interesting part

- **Mercury**: $e = 0.206$, $i = 7.0^\circ$.
- **Pluto**: $e = 0.25$, $i = 17^\circ$.

Both point to a more complicated dynamical history than "formed here, stayed here". Pluto's case we solve later today, with resonances.



5. The two-body problem and vis-viva

Comet 1P/Halley, 21 March 1986. Credit: R. Haefner / ESO, CC BY 4.0

Reducing two bodies to one

Two masses m_1 and m_2 interacting through gravity alone. In the centre-of-mass frame this becomes a *single* body of **reduced mass**

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

moving in the field of a fixed central mass $M = m_1 + m_2$.

The total orbital energy is then

$$E = \frac{1}{2}\mu v^2 - \frac{G m_1 m_2}{r}$$

- For a planet around a star, $m_2 \ll m_1$, so $\mu \to m_2$ and $M \to m_1$: the familiar one-body picture.
- The reduction is exact, not an approximation. It is what makes the two-body problem solvable in closed form, and the three-body problem not.

The central result: energy depends only on a

$$\boxed{E = -\frac{Gm_1 m_2}{2a}}$$

- No e anywhere on the right-hand side.
- Two orbits with the same semi-major axis have the **same total energy**, however different their shapes.
- A circular orbit and a wildly eccentric one with the same a are energetically *equivalent*: eccentricity only redistributes kinetic and potential energy *along* the orbit.

This is why a is the natural coordinate for everything dynamical: migration changes energy, resonance and tides trade e against it.

The vis-viva equation

Combining the total energy with the instantaneous kinetic plus potential energy gives *vis viva*, Latin for “living force”:

$$\boxed{v^2 = G(m_1 + m_2)\left(\frac{2}{r} - \frac{1}{a}\right)}$$

For $m_2 \ll m_1$: $v^2 = GM(2/r - 1/a)$, the speed at *any* point of *any* orbit.

Two special cases worth memorising:

- **Circular orbit** ($r = a$): $v_{\mathrm{circ}} = \sqrt{GM/a}$
- **Escape** ($a \to \infty$, $E = 0$): $v_{\mathrm{esc}} = \sqrt{2GM/r}$

Escape velocity is exactly $\sqrt{2}$ times the circular velocity at the same distance. Worth remembering, and worth checking on the exam.

6. Blackboard derivation: the vis-viva equation

Comet 1P/Halley’s nucleus, Giotto HMC (1986). Credit: ESA / MPAe Lindau

Setup

A planet of mass m orbiting a star of mass M, with $m \ll M$. Orbit has semi-major axis a and eccentricity e. At any moment the planet is at distance r and moves with speed v.

Total mechanical energy is conserved:

$$E = \frac{1}{2}mv^2 - \frac{GMm}{r} = \text{constant}$$

Goal: express E in terms of a alone, then solve for $v(r)$.

The trick throughout is to evaluate the conserved quantities at the two points where the geometry is simplest.

Step 1: evaluate the energy at perihelion and aphelion

At those two points the distances are fixed by the geometry of the ellipse:

$$r_p = a(1 - e), \qquad r_a = a(1 + e)$$

Energy conservation between them:

$$\frac{1}{2}mv_p^2 - \frac{GMm}{a(1 - e)} = \frac{1}{2}mv_a^2 - \frac{GMm}{a(1 + e)}$$

Two unknown speeds, one equation. We need a second relation between v_p and v_a.

Step 2: use angular momentum

At perihelion and aphelion, and *only* there, the velocity is purely tangential. So the angular momentum is just *mvr*:

$$L = m\, v_p\, r_p = m\, v_a\, r_a \quad \Longrightarrow \quad v_p\, a(1 - e) = v_a\, a(1 + e)$$

$$\frac{v_p}{v_a} = \frac{1 + e}{1 - e}$$

This is Kepler's second law doing the work: equal areas in equal times *is* conservation of angular momentum.

Step 3: eliminate v_p

Substitute $v_p = v_a \frac{1+e}{1-e}$ from Step 2 into the energy equality of Step 1 (divided through by m), and collect kinetic left, potential right:

$$\frac{1}{2}v_a^2\left[\frac{(1+e)^2-(1-e)^2}{(1-e)^2}\right] = \frac{GM}{a}\left[\frac{1}{1-e} - \frac{1}{1+e}\right]$$

Both brackets collapse: $(1+e)^2-(1-e)^2 = 4e$ and $\frac{1}{1-e} - \frac{1}{1+e} = \frac{2e}{(1-e)(1+e)}$:

$$\frac{2e\,v_a^2}{(1-e)^2} = \frac{GM}{a}\,\frac{2e}{(1-e)(1+e)} \quad\Longrightarrow\quad v_a^2 = \frac{GM}{a}\,\frac{1-e}{1+e}$$

(Taking $e \neq 0$; for $e = 0$ the apsides coincide, the orbit is circular, and the L1 force balance gives the result of Step 4 directly.)

Step 3, continued: the total energy

Insert v_a^2 into the energy evaluated at aphelion:

$$\frac{E}{m} = \frac{1}{2}v_a^2 - \frac{GM}{a(1+e)} = \frac{GM}{2a}\frac{1-e}{1+e} - \frac{GM}{a(1+e)}$$

$$= \frac{GM}{a(1+e)}\left[\frac{1-e}{2} - 1\right] = -\frac{GM}{a(1+e)}\frac{1+e}{2}$$

$$\boxed{E = -\frac{GMm}{2a}}$$

- The eccentricity has **cancelled**.
- $E < 0$ for any bound orbit, and $E \to 0$ as $a \to \infty$.

Step 4: write the vis-viva equation

Set the total energy equal to the instantaneous kinetic plus potential energy at an arbitrary point:

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = -\frac{GMm}{2a}$$

Divide through by $m/2$ and rearrange:

$$\boxed{v^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right)}$$

This generalises L1: setting gravity equal to the centripetal acceleration gave $v^2 = GM/r$, which is vis-viva for the special case $r = a$.

Application 1: Earth

For Earth, a = 1 AU = 1.496×10^{11} m and e = 0.0167:

Point	Distance r	Speed
Perihelion	$a(1 - e) = 1.471 \times 10^{11}$ m	$v_p = 30.29$ km s^{-1}
Aphelion	$a(1 + e) = 1.521 \times 10^{11}$ m	$v_a = 29.29$ km s^{-1}

- Variation of only about ±0.5 km s^{-1} around the mean, because the orbit is nearly circular.
- Mean orbital speed $v \approx 2\pi a/P \approx 29.8$ km s^{-1}.

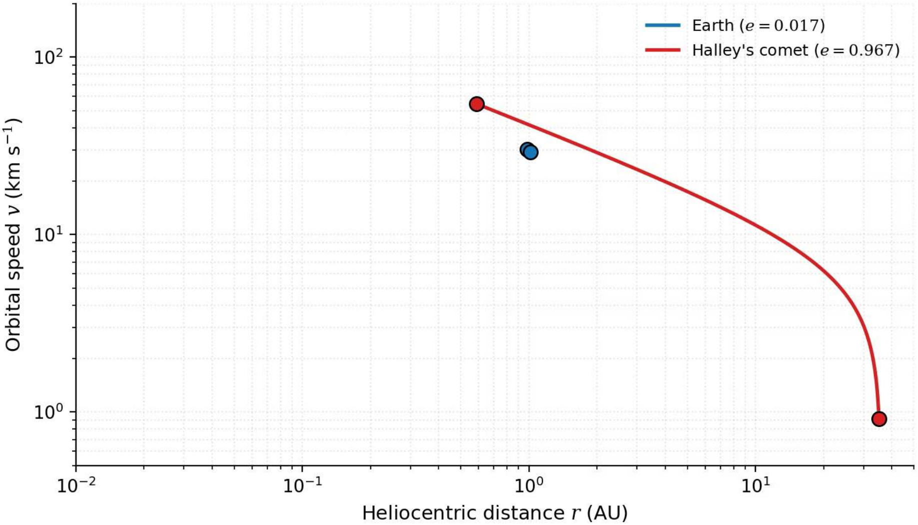
Application 2: Halley's comet

Same equation, an orbit as different as the solar system provides: a = 17.83 AU, e = 0.967.

Point	Distance r	Speed
Perihelion	$a(1 - e) = 0.588$ AU	$v_p = 54.5$ km s^{-1}
Aphelion	$a(1 + e) = 35.1$ AU	$v_a = 0.91$ km s^{-1}

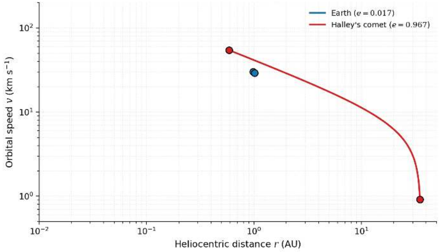
Halley crosses the inner solar system at nearly twice Earth's speed, then crawls beyond Neptune's orbit at under 1 km s^{-1}.

A factor ~ 60 in speed on a single orbit, from one equation and one number: the eccentricity.



Orbital speed $v(r)$ from the vis-viva equation, for Earth and for Halley's comet. Course-original figure.

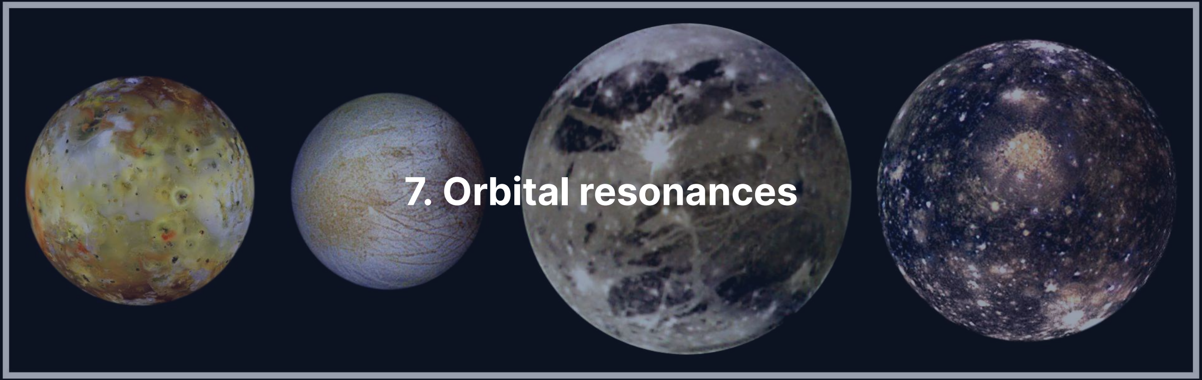
Reading the vis-viva curves



Speed against distance for the two orbits just computed. Filled circles mark perihelion and aphelion.

- **Earth**: a short, almost flat segment near 30 km s^{-1}. Nearly circular means nearly constant speed.
- **Halley**: a steep curve spanning 54.5 km s^{-1} down to 0.91 km s^{-1}.
- Both curves obey the *same* equation with the *same* GM. Only a and e differ.
- The curves never cross: at any shared r, v^2 differs by the *constant* $GM(1/a_{\oplus} - 1/a_{\mathrm{H}})$, so the larger-$a$ orbit is faster. Passing Earth's orbit, Halley moves at 41.5 km s^{-1} against Earth's 29.8.

7. Orbital resonances



The Galilean moons: Io, Europa, Ganymede, Callisto. Credit: NASA/JPL

Mean-motion resonances

When two orbital periods stand in a ratio of **small integers**, the bodies are in a **mean-motion resonance** (MMR).

- A p:q resonance (with $p > q$) means the inner body completes p orbits while the outer completes q.
- Conjunctions then always happen at the *same orbital phases*.
- So the mutual gravitational kicks **add coherently** over many orbits, instead of averaging to zero.

Resonance is a lever: a tiny force applied always at the same phase produces a large cumulative effect. Depending on geometry it can either protect an orbit or destroy it.

Murray & Dermott (1999).

Resonances are everywhere in the solar system

Resonance	Bodies	What it does
3:2	Pluto–Neptune	Pluto completes 2 orbits per 3 of Neptune
Near 5:2	Jupiter–Saturn	"Great inequality": long-period perturbations
3:1, 5:2, 7:3, 2:1	Asteroid belt	Kirkwood gaps, cleared of asteroids (L12)
1:1	Jupiter Trojans	Co-orbital swarms at the L4 and L5 points

- **Protective**: the 3:2 with Neptune is why Pluto can cross Neptune's orbit and never meet it.
- **Destructive**: the Kirkwood gaps are resonances that *pump eccentricity* until the asteroid is removed.

Same physics, opposite outcome, decided by the geometry of the encounter.

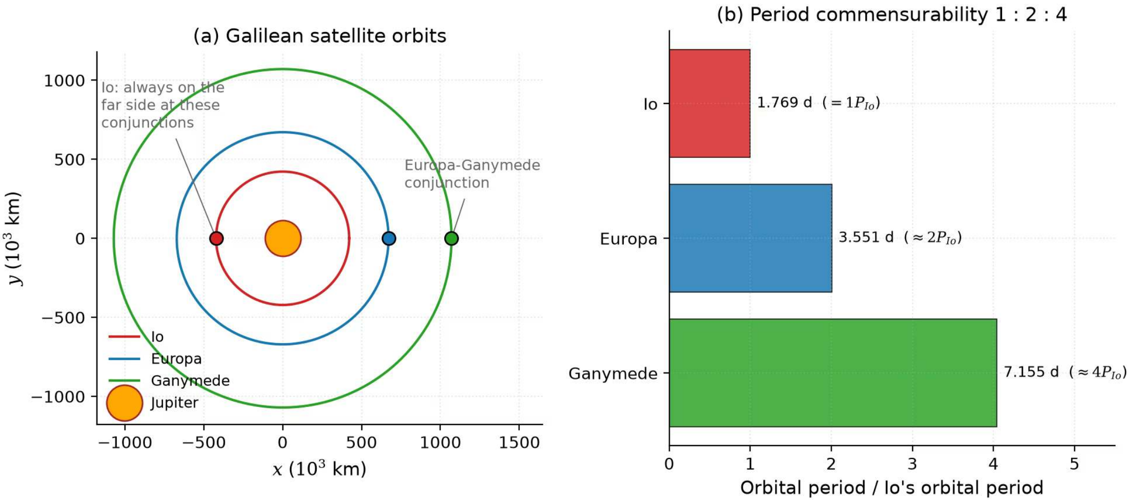
The Laplace resonance

Jupiter's three inner Galilean moons, the most famous resonance in the solar system.

Moon	Period (days)	Ratio to Io
Io	1.769	1
Europa	3.551	≈ 2
Ganymede	7.155	≈ 4

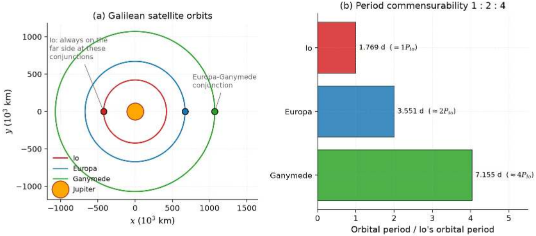
- Periods in a $1:2:4$ ratio, equivalently mean motions $n_{\mathrm{Io}} : n_{\mathrm{Eur}} : n_{\mathrm{Gan}} = 4:2:1$.
- A genuine **three-body** resonance, maintained by tidal interaction with Jupiter.
- Its key consequence: it **forces non-zero eccentricities**. Io's is pinned at $e \approx 0.004$.

Peale, Cassen & Reynolds (1979).



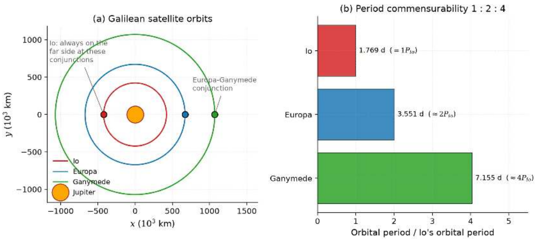
The Laplace resonance of Io, Europa, and Ganymede. Course-original figure.

Reading the Laplace resonance



- **Panel (a)**: the three orbits to scale, drawn at a Europa–Ganymede conjunction. The resonance locks the phases so that Io is always on the *opposite side of Jupiter* at those conjunctions.
- Consequence, and a common misconception to avoid: a **triple conjunction never occurs**. The three moons are never lined up together.

Reading the Laplace resonance



- **Panel (b)**: the periods in their 1 : 2 : 4 ratio. The pattern of successive conjunctions repeats every Ganymede period, that is, every four Io orbits.
- The periodic kicks arrive always at the same phase, which is what sustains Io's $e \approx 0.004$ against tidal damping.

An eccentricity of 0.004 sounds negligible. It is the reason Io is the most volcanically active body in the solar system.

Io: orbital dynamics with thermal consequences

Credit: NASA/JPL/University of Arizona (Galileo)

Half-disk view from Galileo, with an active volcanic plume rising in profile against the dark sky at the limb.

- The mottled, sulphur-rich surface is resurfaced on geological timescales.
- Resonance-forced eccentricity feeds continuous tidal dissipation in the interior.
- Remove the resonance and Io circularises, stops flexing, and goes geologically dead.

We quantify the heating in L3.

8. Tides, locking, and the Roche limit

Saturn backlit, Cassini (2013). Credit: NASA/JPL-Caltech/Space Science Institute

The tidal force

A tidal force arises whenever a gravitational field **varies across the extent** of an extended body.

For a moon of radius R_s at distance d from a planet of mass M_p, the acceleration difference between the moon's centre and its near side is

$$\Delta a_{\mathrm{tidal}} \approx \frac{2GM_pR_s}{d^3}$$

- The near side is pulled harder than the centre, the far side less hard: the body is **stretched** along the line to the planet.
- Note the exponent: $1/d^3$, not $1/d^2$. Tides fall off much faster than gravity itself.

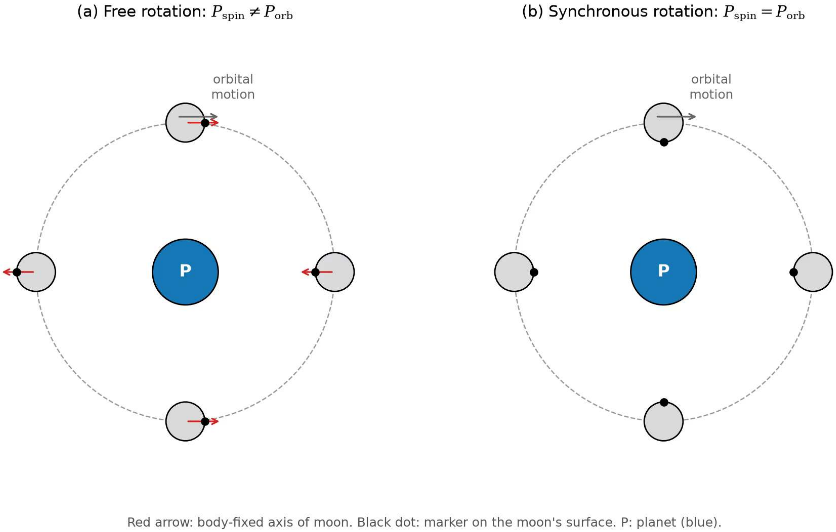
The steep distance dependence is why tides are irrelevant for most of a planetary system and utterly dominant close in.

Bulges, torques, and heat

The tidal force raises **two bulges**, one facing the perturber and one on the opposite side.

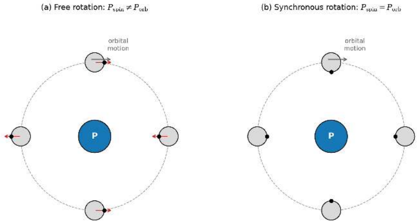
1. If the body rotates at a different rate than it orbits, which is the generic starting condition, the bulges are **carried away** from the line joining the two bodies.
2. That misalignment gives the perturber a lever arm: a **torque** that exchanges angular momentum between spin and orbit.
3. Continually deforming the body **dissipates energy as heat**: **tidal heating**.

Tidal heating powers Io’s volcanism and may maintain the subsurface oceans of Europa and Enceladus (L3, L11).



Free versus synchronous rotation. Course-original schematic.

Tidal locking

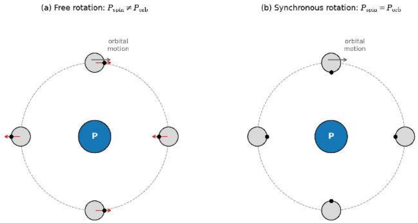


Red arrow: body-fixed axis of moon. Black dot: marker on the moon's surface. P: planet (blue).

Over time the tidal torque slows the faster-spinning body until its rotation period equals its orbital period: **synchronous rotation**.

- **Panel (a)**, free rotation: spin and orbital periods differ, so the moon shows different faces to the planet (P) over one orbit. The black surface marker and the body-fixed axis (red arrow) decouple from the direction to P.
- **Panel (b)**, **synchronous rotation**: the periods are equal, so the marker, and with it the tidal bulge, always faces the planet.

Tidal locking



Red arrow: body-fixed axis of moon. Black dot: marker on the moon's surface. P: planet (blue).

- With the bulge aligned, the torque vanishes and the configuration is stable.
- The Moon reached this state through billions of years of dissipation. All four Galilean moons are locked to Jupiter; most large moons in the solar system are synchronous rotators.

Murray & Dermott (1999).

Locked does not mean cold

A common and consequential misconception: that synchronous rotation switches tidal dissipation off.

- On a **circular** orbit it essentially does. Bulge aligned, nothing flexes, no heat.
- On an **eccentric** orbit it does not. Distance and apparent bulge direction both vary over each orbit, so a locked body **keeps flexing**.

Exactly Io's situation: locked to Jupiter, but with $e \approx 0.004$ pinned by the Laplace resonance.

How fast does locking happen?

The timescale scales as d^6/M_p^2: steeply with distance, inversely with the square of the primary's mass. Close-in moons of massive planets lock almost immediately; distant, small companions may never lock at all.

The Roche limit

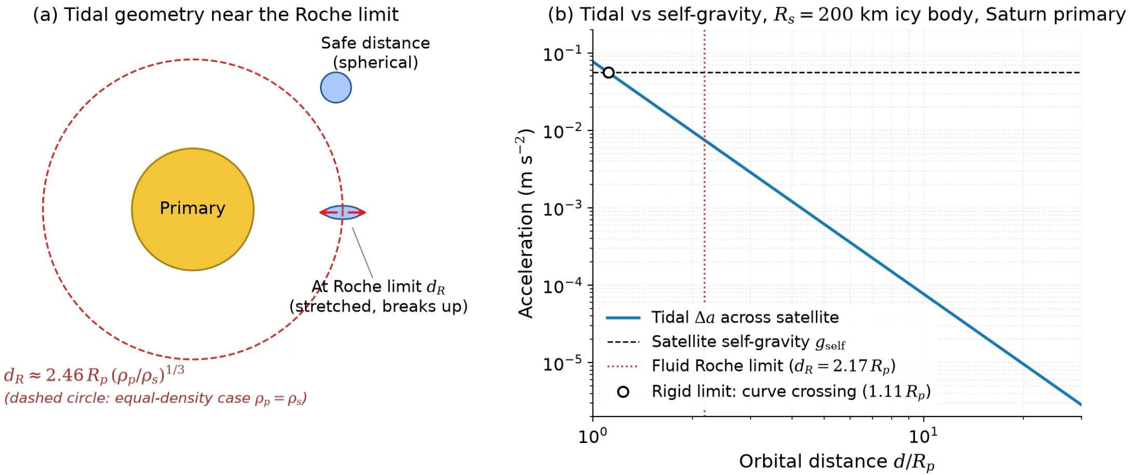
There is a minimum distance at which a self-gravitating body survives the tide of a larger one. Inside it, tides beat self-gravity and the body is torn apart.

For a *fluid* satellite of density ρ_s orbiting a planet of radius R_p and density ρ_p:

$$d_R \approx 2.46\, R_p \left(\frac{\rho_p}{\rho_s}\right)^{1/3}$$

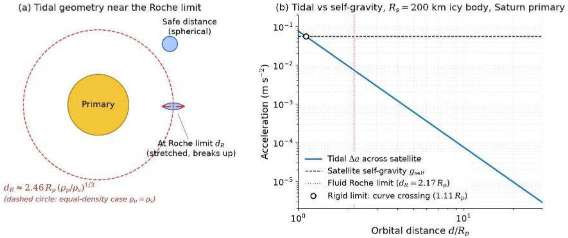
- Note what does *not* appear: the satellite's **size**. Only the densities and the primary's radius matter.
- The numerical coefficient depends on the assumed satellite rheology, which is the subject of the next slide.

We derive this in full in L11.



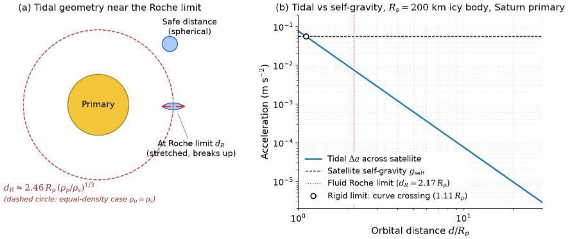
Roche limit geometry, and the tidal-versus-self-gravity crossing for a 200 km icy body at Saturn. Course-original figure.

Reading the Roche geometry



- **Panel (a)**: a fluid satellite is rounded at a safe distance. At d_R the differential tidal acceleration between its centre and its surface exceeds its own self-gravity, and it is stretched and torn apart along the line to the primary.
- **Panel (b)**: tidal acceleration compared with surface self-gravity for a 200 km icy body, against distance from Saturn in units of R_p.

Reading the Roche geometry



- The simple **crossing** of the two curves gives the *rigid* limit, $d_R^{\rm rigid} \approx 1.11\,R_p$ for ρ_s = 1000 kg m^{-3}.
- The textbook *fluid* limit, $d_R \approx 2.17\,R_p$ (red dotted), lies further out because it adds the tidal **deformation** of the satellite, which weakens it further.

Two limits, a factor of two apart. Which one applies depends on whether the body can hold its shape, so real bodies fall between them.

Evaluating the Roche limit at Saturn

With R_p = 58,232 km, ρ_p = 687 kg m^{-3}, and an icy satellite $\rho_s \approx$ 1000 kg m^{-3}:

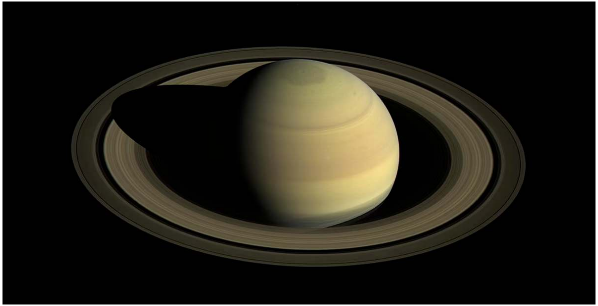
$$d_R \approx 2.46 \times 58{,}232 \text{ km} \times \left(\frac{687}{1000}\right)^{1/3} \approx 126{,}000 \text{ km} \approx 2.17\, R_p$$

Compare with the actual ring system:

- Main rings (D through A): ~ 67,000 km to 137,000 km from Saturn’s centre.
- The narrow F ring lies just beyond, at ~ 140,000 km.
- The dense **B and C rings sit well inside** the fluid limit, as expected.
- But **much of the A ring’s width lies outside** it.

Two reasons the simple number is not the whole story: the A ring’s sharp outer edge is held by a **7:6 resonance with Janus**, and the particles are **porous**, so their density is below solid ice, which pushes the effective limit outward.

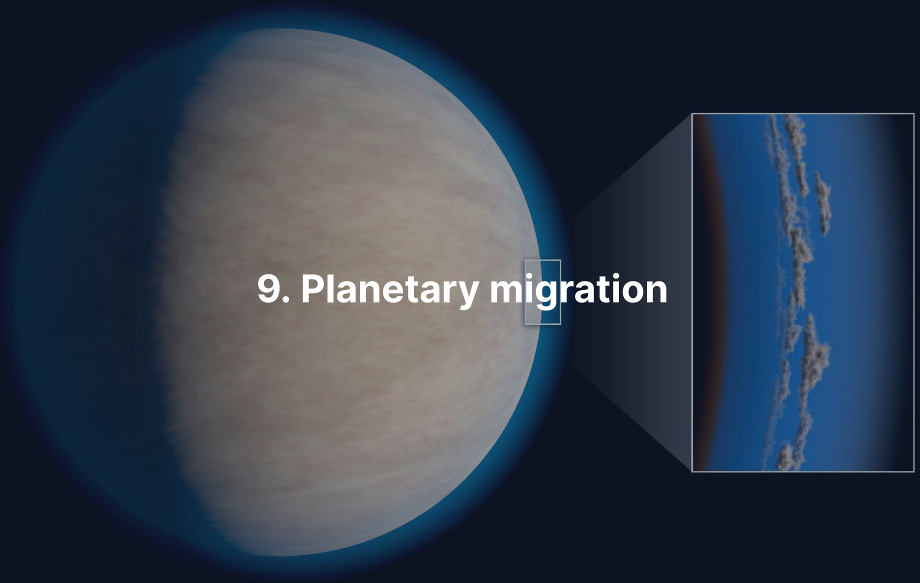
Saturn’s rings



Credit: NASA/JPL-Caltech/Space Science Institute (PIA21046)

- Cassini wide-angle view, April 2016.
- Countless cm- to m-sized icy particles.
- They cannot coalesce: Saturn’s tide beats their mutual self-gravity.
- A ring system is a **moon that is not allowed to assemble**.
- Run in reverse, this is why large moons avoid very tight orbits.

9. Planetary migration



Hot Jupiter, artist’s concept. Credit: ESO/L. Calçada

Planets do not stay where they form

A planet embedded in a gaseous disk exchanges angular momentum with it, through two distinct channels.

- **Lindblad resonances.** The planet excites spiral density waves at the radii where the disk's orbital frequency differs from the planet's by $\pm\kappa/m$, with κ the local epicyclic frequency and m a positive integer. The waves carry angular momentum away.
- **Corotation resonances.** Material co-orbiting with the planet exchanges angular momentum along **horseshoe-shaped streamlines**.

The *net* torque decides whether the planet moves in or out, and how fast.

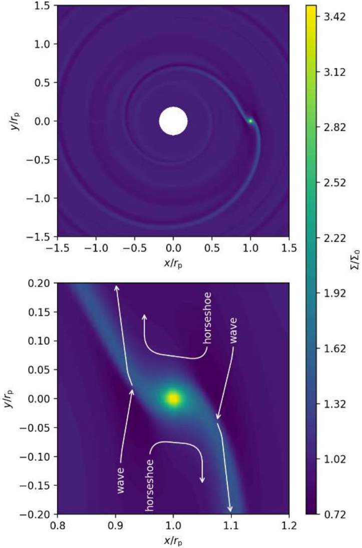
Migration is not an exotic add-on. Any planet massive enough to perturb the disk is being pushed by it.

Paardekooper et al. (2023).

Three types of migration

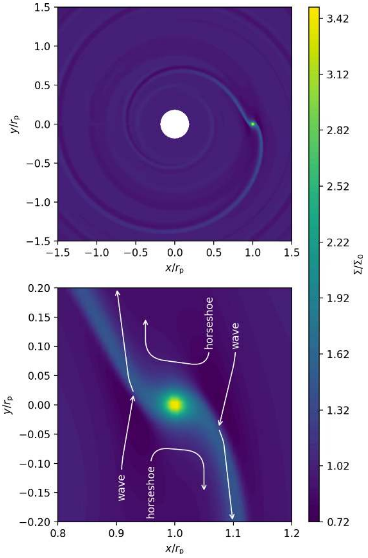
- **Type I.** A low-mass planet (below the gap-opening threshold, a few tens of Earth masses) stays fully embedded and migrates on the imbalance between inner and outer Lindblad torques. In simple models the outer torque slightly wins: *inward* migration on ~ 10^5 yr.
- **Type II.** A massive planet, roughly Jupiter-mass and up, **opens a gap**: an annulus cleared of gas. It then migrates on the slower viscous timescale, ~ 10^5–10^6 yr, as the disk drains onto the star.
- **Type III.** In massive disks with a sharp density gradient near the planet, a runaway sets in when the **co-orbital mass deficit** approaches the planet's mass. Very fast, either direction.

The Type I problem: 10^5 yr is uncomfortably short against a ~ 3 Myr disk lifetime. Taken literally, cores fall into the star before they finish growing; extra physics (thermal effects, magnetic fields, disk structure) must slow or reverse it.



Disk surface-density response to an embedded Neptune-mass planet. Adapted from Paardekooper et al. (2023), Fig. 2.

Reading the disk-flow figure



A planet of mass ratio $q = 10^{-4}$, Neptune-like, on a circular orbit.

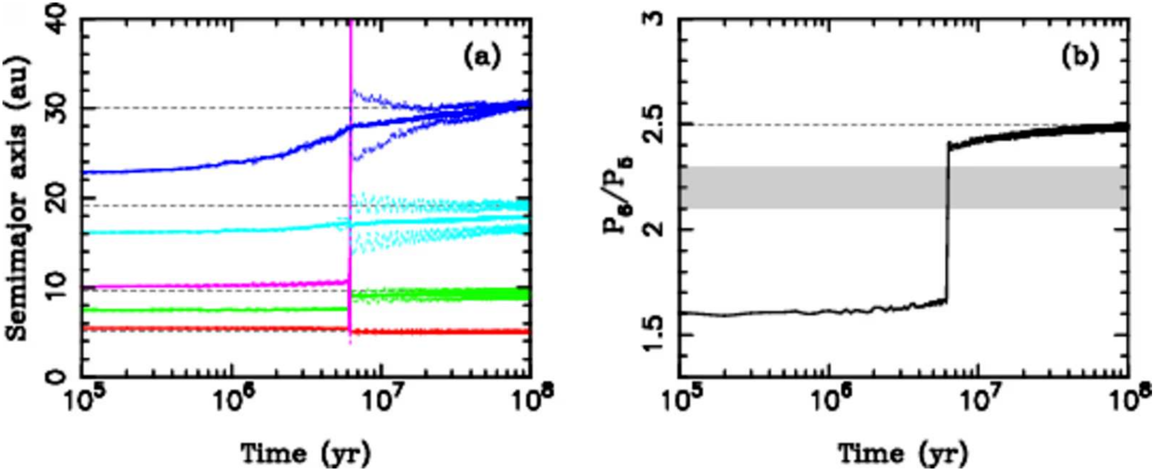
- **Top panel**: the planet launches a one-armed **spiral density wave**. Colour is surface density relative to unperturbed.
- Note what is absent: **no gap**. At this mass the planet perturbs the disk without clearing it, which is the definition of the Type I regime.
- **Bottom panel**: zoom on the co-orbital region, with schematic streamlines showing the **horseshoe trajectories** and the flows feeding the spiral wave.
- The imbalance between inner and outer wave torques, plus the corotation torque from the horseshoe region, sets both the direction and the speed.

The Nice model

Migration continues after the gas goes: the giants scatter a remnant **planetesimal disk**.

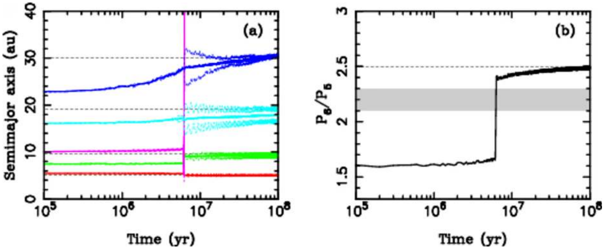
- The giants formed **more compact**, the outermost ice giant well inside Neptune's present 30 AU orbit, until a dynamical instability rearranged the system.
- **Original version**: triggered when Jupiter and Saturn slowly crossed their mutual **2:1** resonance.
- **Modern version**: the belt's structure and the near-circular terrestrial orbits require the Jupiter–Saturn period ratio to change **discontinuously**, most naturally if a **fifth giant planet** was scattered off Jupiter and ejected: *jumping Jupiter*.
- Either way: Uranus and Neptune scattered outward, the primordial Kuiper Belt disrupted, and possibly an impact spike inside (the "Late Heavy Bombardment", still debated).

Tsiganis et al. (2005); Nesvorný (2018).



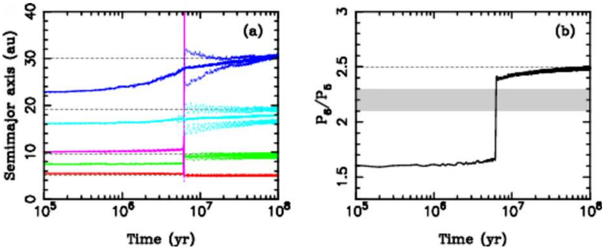
A five-planet “jumping-Jupiter” instability. Reproduced from Nesvorný (2018), Fig. 3; simulation from Nesvorný & Morbidelli (2012).

Reading the jumping-Jupiter simulation



- **Panel (a)**: semi-major axes vs time for five giants in a compact resonant chain, with an extra ice giant (magenta) between Saturn and Uranus. Dashed lines mark today's values.
- At the instability, ~ 6 Myr in, the extra ice giant is scattered off Jupiter and **ejected**; Uranus and Neptune move outward onto their present orbits.

Reading the jumping-Jupiter simulation



- **Panel (b)**: the Saturn/Jupiter period ratio starts near the 3:2 value of 1.5, drifts up through planetesimal-driven migration, then **jumps** from ≈ 1.7 to ≈ 2.4 in under 10^5 yr, approaching today's 2.49.
- **The shaded band**, $2.1 \lesssim P_{\mathrm{Sat}}/P_{\mathrm{Jup}} \lesssim 2.3$: crossing it *slowly* would activate secular resonances that excite the belt and the terrestrial orbits.

Avoiding that band is the central argument for jumping Jupiter over the smooth 2:1 crossing; the evidence is the solar system we have.

The Grand Tack

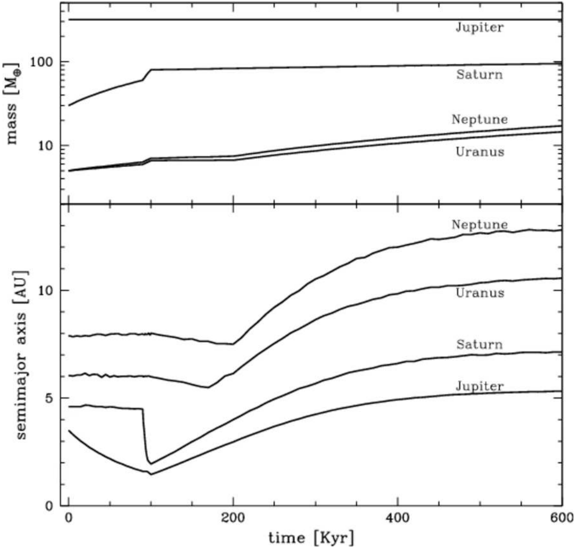
A proposal for what Jupiter did *while* the gas disk was still present:

1. Jupiter migrates **inward** to ~ 1.5 AU by Type II migration.
2. Saturn, migrating faster, catches up and is captured into the **3:2 resonance** with Jupiter.
3. Their combined interaction with the disk reverses the torque, and both planets move **outward**: the "tack", as in sailing.

The inward excursion would have depleted solids in the inner disk, which could explain **Mars's small mass** and the belt's compositional structure.

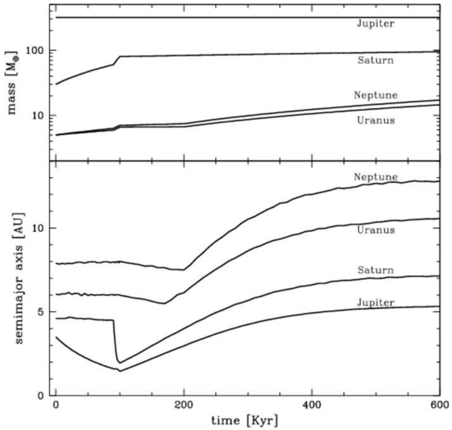
Read it as a hypothesis with consequences, not a settled result: the migration in the reference simulation is an *imposed prescription*, not an outcome of the model.

Walsh et al. (2011).



The Grand Tack scenario: giant-planet growth and the imposed migration tracks. Reproduced from Walsh et al. (2011), Fig. 1.

Reading the Grand Tack tracks



- **Top panel**: mass growth of the four giant planets during the disk phase.
- **Bottom panel**: semi-major axis against time, each curve labelled with its planet.
- Follow Jupiter: inward to ~ 1.5 AU, then the **reversal** once Saturn catches up and the 3:2 resonance locks them together, then outward.
- **Important caveat**: these tracks are the migration *prescription imposed* on the planets in the reference simulation, following the inward-then-outward path found in hydrodynamical studies. The figure shows the scenario, not a derivation of it.

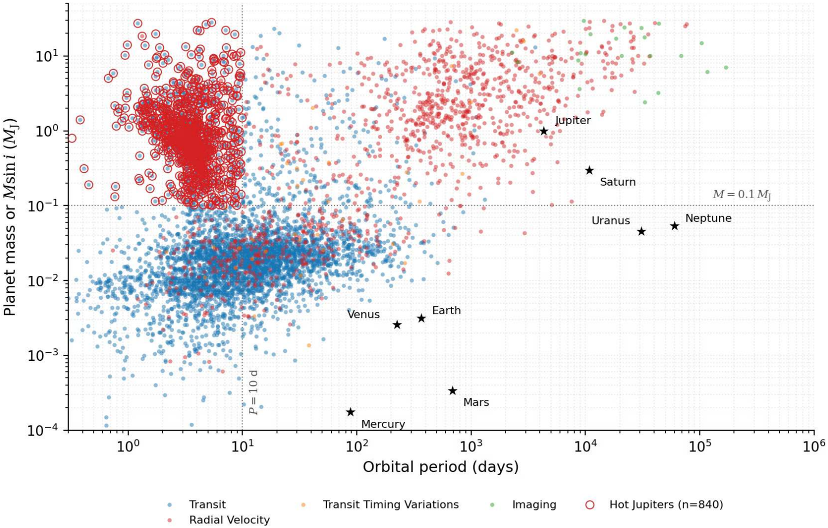
The observational case for migration

The strongest evidence is a population that cannot be explained any other way: **hot Jupiters**.

- Gas giants with orbital periods of a few days, $a \lesssim 0.1$ AU.
- They **cannot have formed in situ**: there is not enough material that close in, and it is far too hot for solids.
- So they formed further out and moved inward, by *either*
  - **disk migration**, smooth and early, or
  - **high-eccentricity migration** followed by tidal circularisation, violent and late.

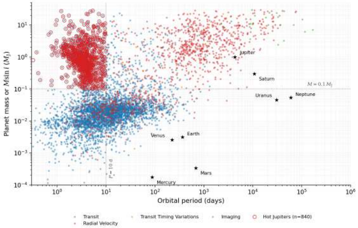
Independently: the gaps ALMA sees in disks such as HL Tau may be the signature of embedded planets already migrating. We return to exoplanet demographics in L13.

Dawson & Johnson (2018).



Confirmed exoplanets in mass–period space. Course-original figure; data: NASA Exoplanet Archive (accessed 2026-08).

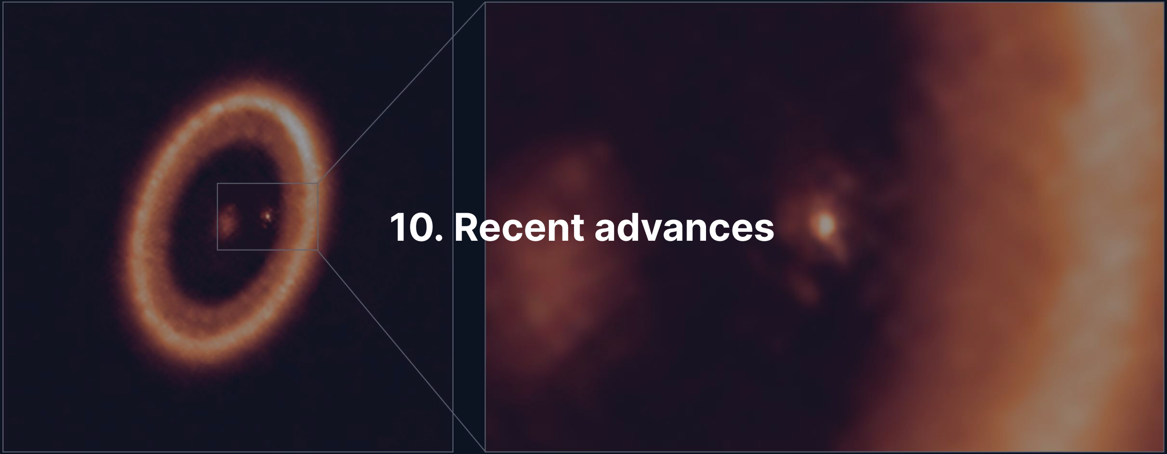
Reading the mass–period diagram



- Every confirmed planet with a measured mass and period, colour-coded by **detection method**. The method boundaries are selection effects, not physics: remember that before reading structure into the plot.
- **Black stars**: the eight Solar System planets, for reference.
- **Red circles**: hot Jupiters, selected here as $M > 0.1\,M_J$ and $P < 10$ d (dotted lines).
- That clump sits where no planet should be able to form. It is the population that made migration unavoidable in the theory.

Our solar system has no hot Jupiter. Whether that makes it unusual is a question we can only ask now that we have the comparison sample (L13).

10. Recent advances



PDS 70, protoplanets in a disk gap. Credit: ALMA (ESO/NAOJ/NRAO)

What high-resolution imaging changed

- DSHARP found rings and gaps in **all 20** of its large, bright target disks, including around stars younger than ~ 1 Myr.
- If embedded planets carve those gaps, then planet formation **begins earlier and proceeds faster** than the classical picture allowed, which is what the pebble-accretion paradigm predicts.
- The honest caveat: ice lines and magnetically structured zones remain viable alternatives for some of the observed structures.

The debate is no longer whether disks are structured. It is what the structure means.

Andrews et al. (2018); Lambrechts & Johansen (2012).

Where the theory stands

The *Protostars and Planets VII* volume consolidates the current synthesis, incorporating the ALMA constraints. Key developments:

- Refined models of the **streaming instability** and the conditions under which planetesimals actually form.
- Improved treatment of **pebble accretion** in evolving, structured disks.
- New constraints on giant-planet formation timescales from **meteorite isotopes**: the inner and outer solar system stayed chemically separated early, which dates Jupiter's core.
- Juno gravity data revealing Jupiter's **dilute core**, which is not the sharp core-envelope boundary the classical model assumed (L8).

Together these point to giant-planet cores growing within the first few Myr, before the gas disperses, although *how early* Jupiter's core in particular had to form is still contested (L12, L14).

Drążkowska et al. (2023); Kruijer et al. (2017); Wahl et al. (2017).

Next time: Lecture 3

This lecture assembled the planets and set their orbits. The next asks what happens *inside* them.

L3 follows the energy:

- Heat delivered by accretion and differentiation.
- Short-lived radionuclides, led by $^{26}\mathrm{Al}$ (half-life 0.72 Myr), that melted even small planetesimals.
- Long-lived isotopes still powering Earth's 47 TW heat flow today.
- Tidal dissipation, whose consequences we just saw on Io.

Then: how that heat *escapes*, by conduction and by convection. The balance between the two decides whether a planet convects, sustains a dynamo, and stays geologically alive.

Summary

1. **Disks are inevitable**: angular momentum forbids direct collapse onto the star, and the disk's temperature structure sets the snow line at 2.5–3.5 AU.
2. **Growth is not incremental**: bouncing, fragmentation, and radial drift stop sticking well short of planetesimals; the streaming instability jumps the gap.
3. **Giant-planet formation is a race** against a ~ 3 Myr disk lifetime, and core accretion only just wins it.
4. **Vis-viva**, $v^2 = GM(2/r - 1/a)$, gives the speed anywhere on any orbit; the energy depends only on a.
5. **Resonances and tides** convert orbital configuration into interior heat: Io's $e \approx 0.004$ is the reason it is the most volcanically active body we know.
6. **Migration is the rule**: hot Jupiters, the Nice model, and the Grand Tack all say that where a planet is now is not where it formed.

Questions?

Next: Lecture 3, Planetary heat & energy transport

Mon 14 Sep, 15:00–17:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 11 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

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